

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix

Source: Our World in Data – Ritchie, Roser & Rosado | Data: Ember (2026); Energy Institute (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse
gas emissions from
fossil fuel combustion

5M+

premature deaths per
year from fossil fuel
air pollution

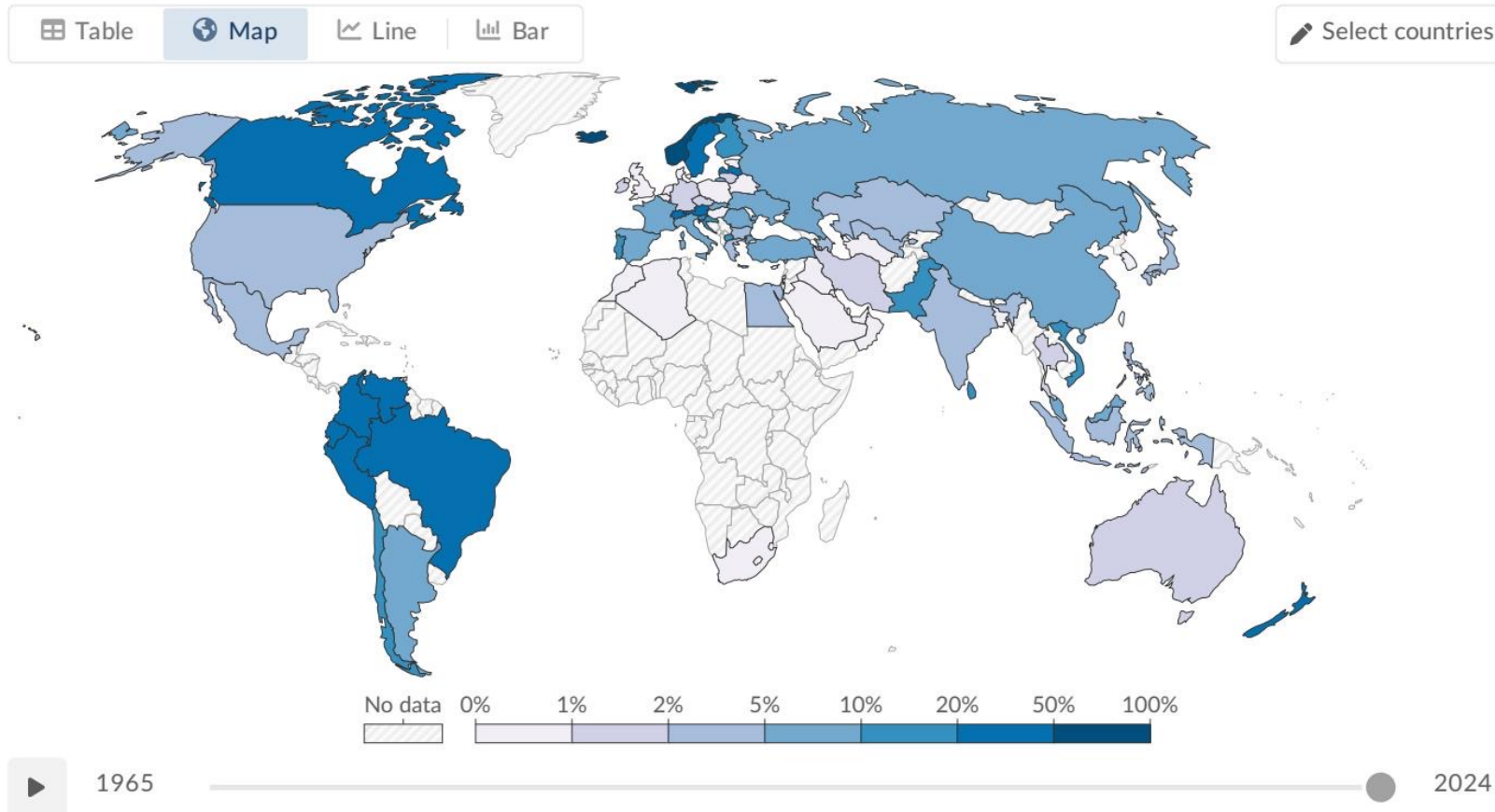
**Since
1750s**

fossil fuels have
dominated the global
energy system

To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies.

Renewables Now Supply ~14% of Global Primary Energy

Share of primary energy from renewables by country, 2024
(substitution method)



~1/7

of global primary energy
is now renewable

Norway, Brazil & New Zealand lead with
high renewable shares

Middle East & parts of Asia remain below
5%

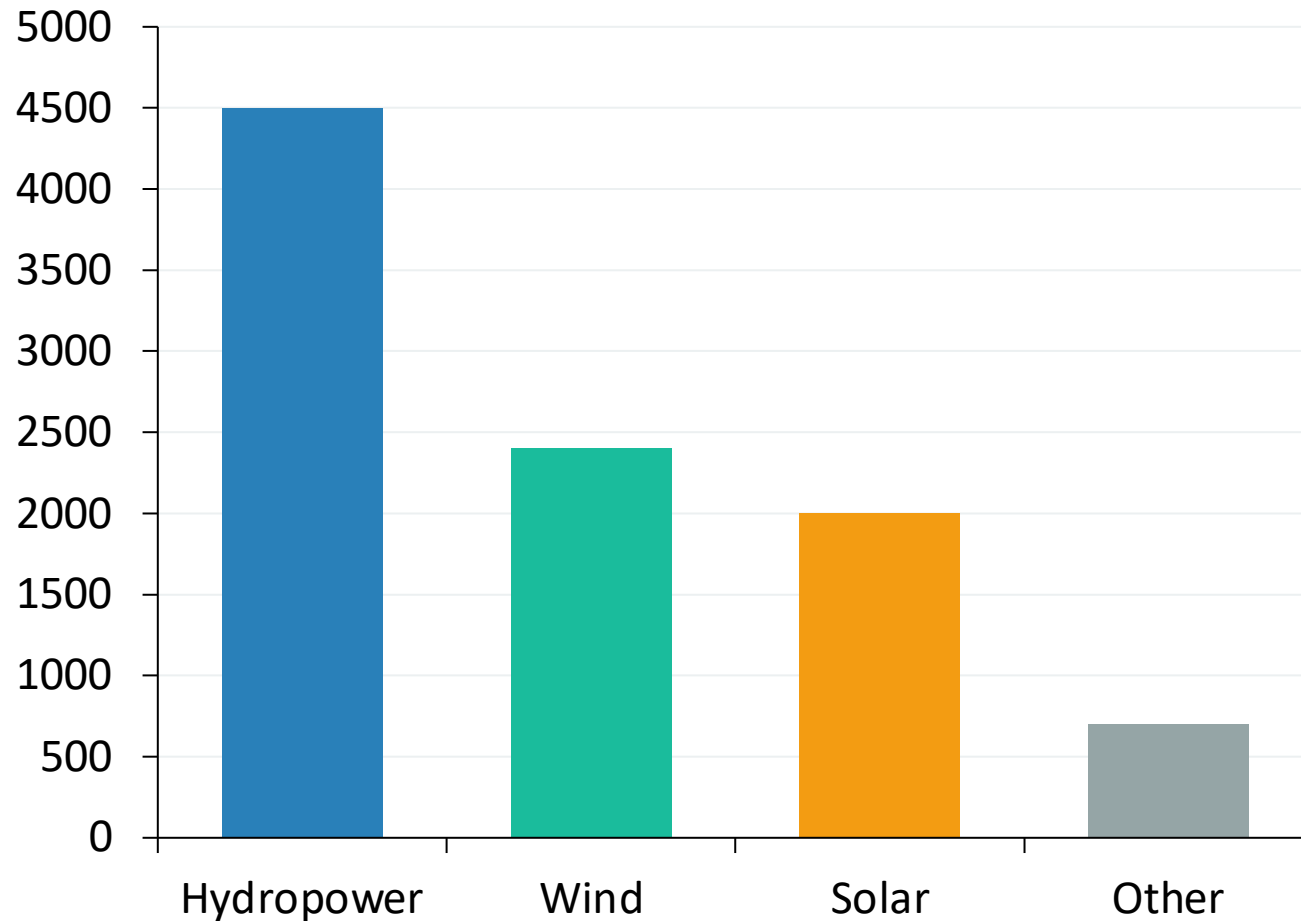
Transport & heating are harder to
decarbonize than electricity

Data source: Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

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Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



Hydropower: ~4,500 TWh

~47% of renewable electricity

Wind: ~2,400 TWh

~25% of renewable electricity

Solar: ~2,000 TWh

~21% of renewable electricity

Other: ~700 TWh

~7% (geothermal, biomass, etc.)

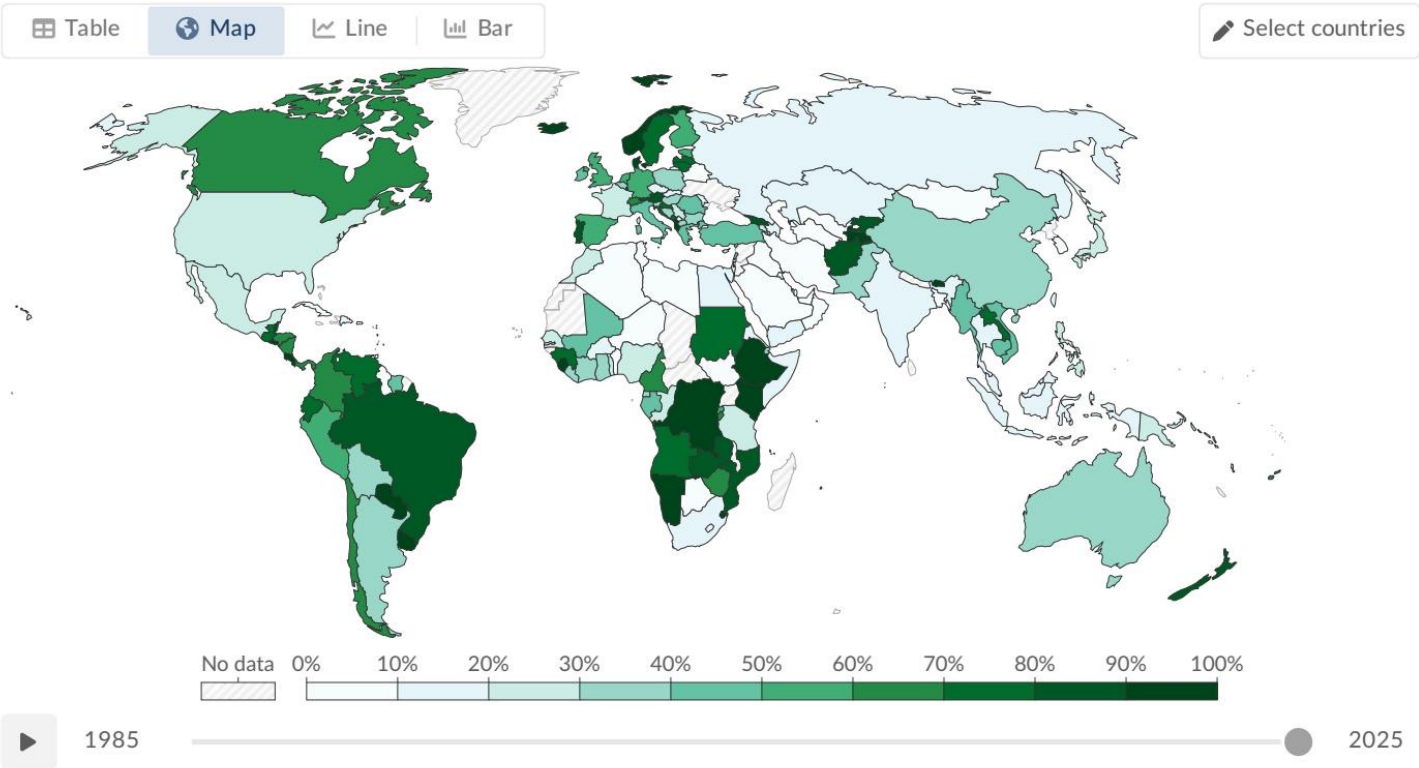
Wind + Solar combined now nearly match Hydro output

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

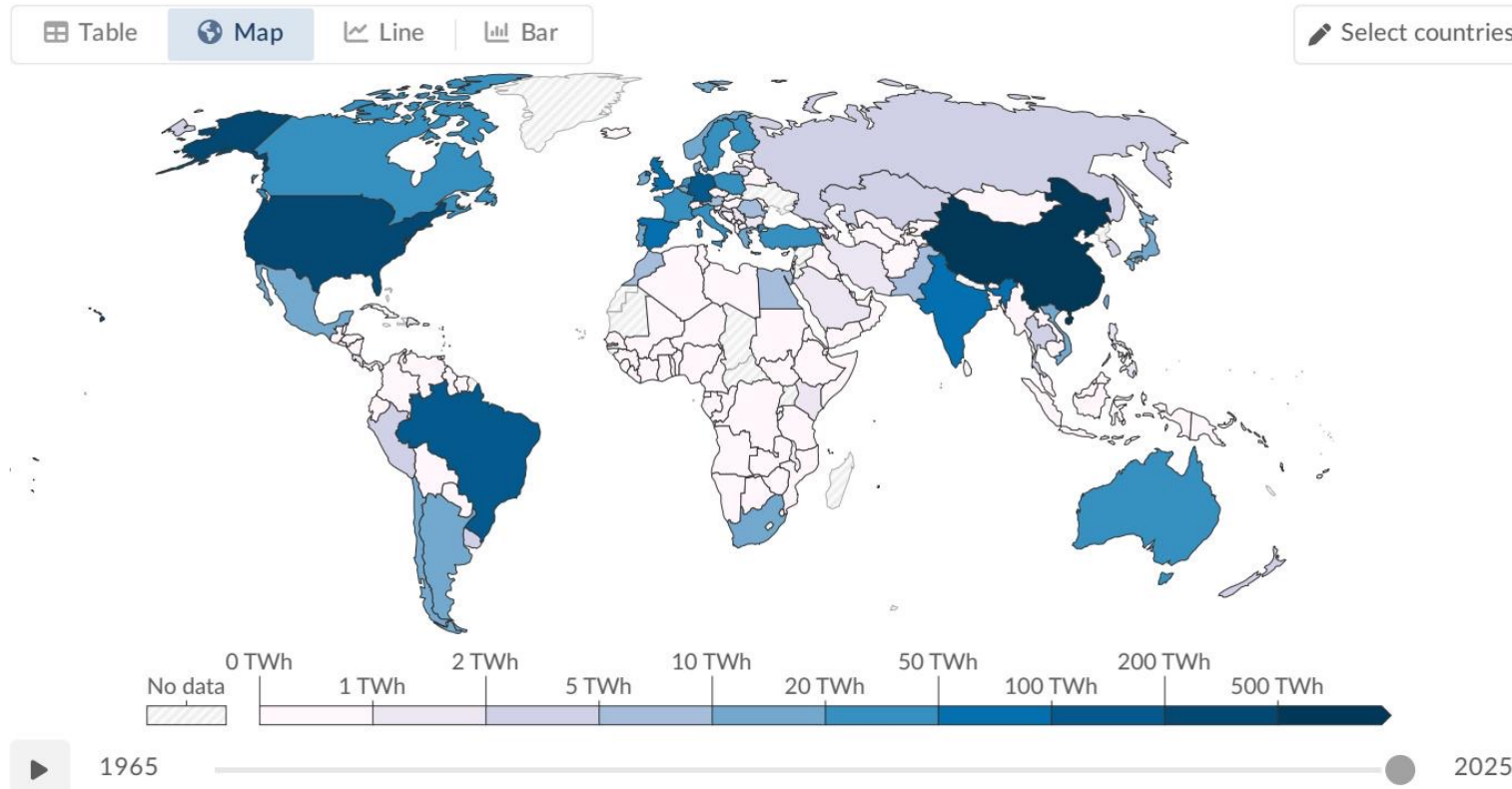


Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
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Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World Average	~30%

Source: Ember (2026), Energy Institute (2025)

Hydropower Remains the Largest Renewable Source at ~50% of Generation



~4,500 TWh

global hydropower generation

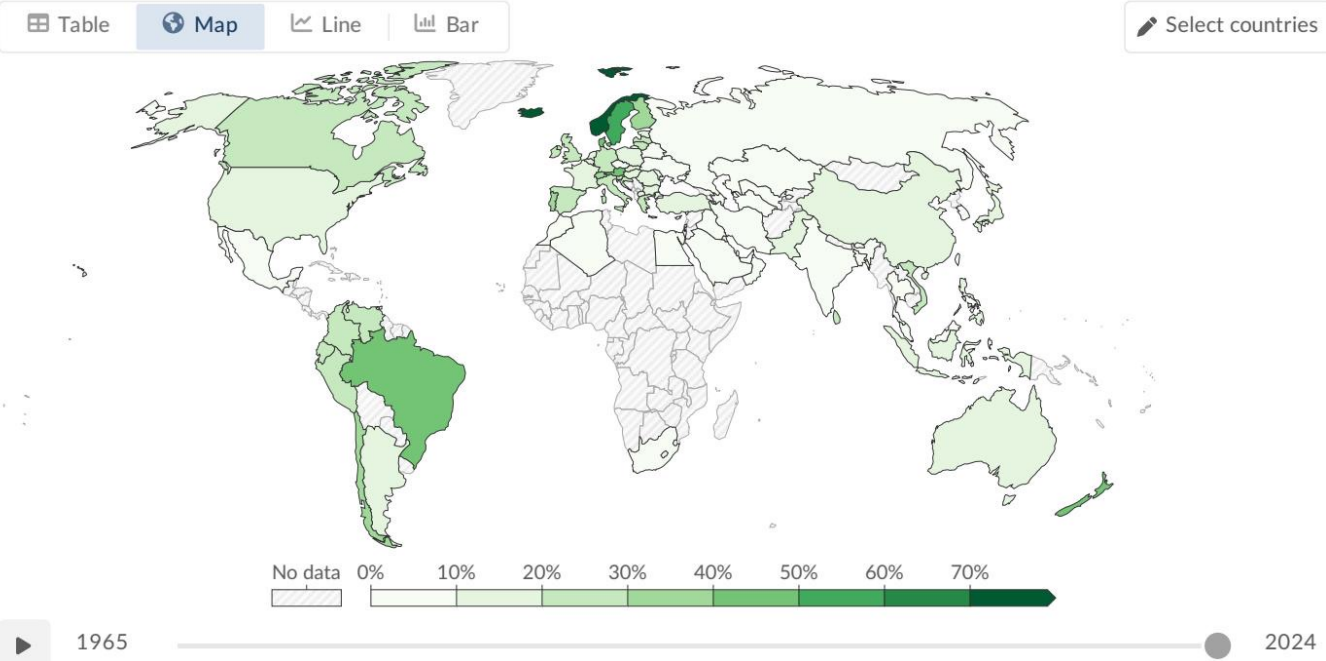
Accounts for ~47% of all renewable electricity

Generating at scale for over a century

Top producers: China, Brazil, Canada

Growth potential limited compared to wind & solar

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind

~2,400 TWh

~25% of renewable
electricity

Major contributor in Europe,
China & US; includes onshore
& offshore

Solar

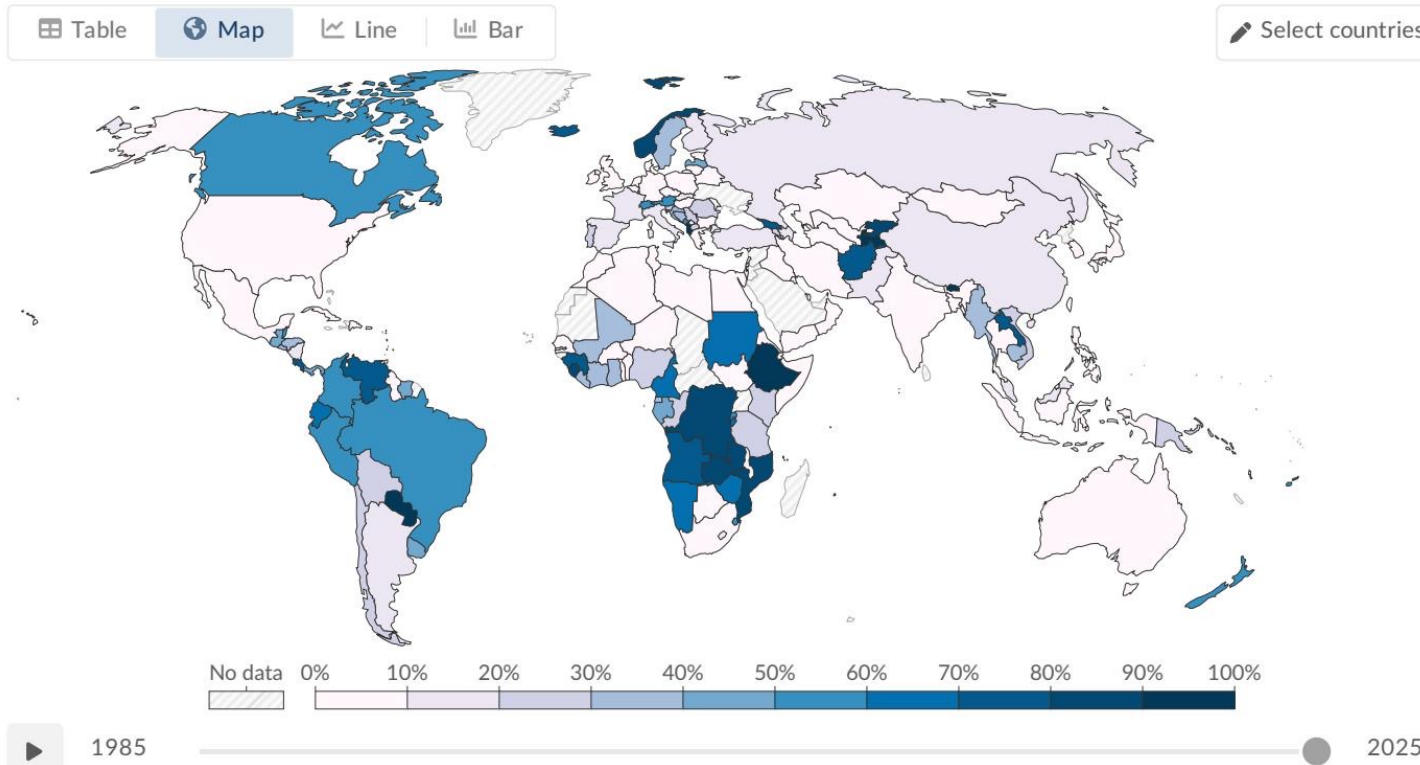
~2,000 TWh

~21% of renewable
electricity

Fastest-growing source;
dramatic growth since 2010
driven by falling costs &
policy

Combined, wind and solar now produce nearly as much electricity as hydropower.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Leaders (65–98% renewable electricity):

Norway, Brazil, Canada – abundant hydropower
Denmark, Germany – heavy wind & solar investment

Laggards (<10% renewable electricity):

Sub-Saharan Africa – minimal renewable infrastructure
Middle East – almost entirely fossil-fuel dependent

The energy transition is highly uneven — geography, investment, and policy all shape outcomes.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

~14% of primary energy & ~30% of electricity now come from renewable sources — a significant milestone but far from sufficient for climate targets.

Hydropower remains dominant generating roughly half of all renewable electricity, but its growth potential is limited.

Wind & solar are the fastest-growing energy technologies in history; combined, they now produce nearly as much electricity as hydro.

Stark regional disparities persist Scandinavia & Latin America exceed 60–90%+, while the Middle East & Sub-Saharan Africa remain below 10%.

Transport & heating lag far behind electricity keeping the overall primary energy share much lower than the electricity share.

The pace of the renewable transition must accelerate dramatically to meet global climate targets.